

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

Mid Semester Examination, Autumn 2023-24

Duration: 2 Hours

Subject: Machine Learning

Subject Code: CS6218

Full Marks: 30

Answer All questions. All parts of a question must be answered at one place. Mere answer without justification will not fetch any mark.

1. (a) Build a linear regression model to find the regression parameters for the following dataset. [6]

x_1	x_2	y
0	1	6
0	0	4
1	0	5
1	1	7

- (b) Define the terms (i). Sensitivity (ii). Over-fitting (iii) k -fold cross validation (iv) AUC [4]
2. (a) Construct a decision tree based on the following data taking gini-impurity index as the splitting criterion. [6]

city	job	sex	class
Bangalore	employed	male	yes
Chennai	student	male	yes
Chennai	employed	male	no
Chennai	unemployed	male	no
Bangalore	employed	female	yes
Bangalore	student	female	no
Chennai	employed	female	yes

- (b) Consider a one dimensional two category classification problem. Suppose the class conditional [4] densities are given by

$$p(x/c_1) = \begin{cases} 0 & \text{if } x < 0 \\ \theta_1 e^{-\theta_1 x} & \text{if } x \geq 0 \end{cases} \quad p(x/c_2) = \begin{cases} 0 & \text{if } x < 0 \\ \theta_2 e^{-\theta_2 x} & \text{if } x \geq 0 \end{cases}$$

Suppose you have collected data for class 1 and stored it in array $D_1 = \{1, 5\}$ and data for class 2 and stored it in array $D_2 = \{3, 6, 9\}$.

- (i) Find the maximum likelihood estimates for parameters θ_1 and θ_2 .
- (ii) Using ML classifier, find decision regions and decision boundaries using the parameters estimated in (i)

3. (a) Find the illness class predicted by the Naive Bayes Classifier for the input vector $X = (Chennai, Female)$ given the following table. [5]

city	gender	illness
Bangalore	Male	Yes
Chennai	Female	Yes
Chennai	Male	Yes
Chennai	Female	Yes
Bangalore	Female	Yes
Bangalore	Male	No
Chennai	Female	No
Chennai	Male	No

- (b) Find one dimensional projection of the following data using PCA. [3]
 $x_1 = (1, 2), x_2 = (2, 3), x_3 = (2, 1)$
- (c) What are the different ways to validate a MLR model? How we measure the correlation between different features? [2]



NITR/CSE/RD/TKM/CS6218/MID/2023.